

# **Type 2 Emergence and First Prospective v1.5 C2 Calibration on a Skincare IL-Gradient Substrate**

AIAS™ Presence Measurement Protocol, v0.20 — Type 2 Quadrant Populated; v1.5 C2 First Prospective Calibrated

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## Abstract

The AIAS™ Presence Measurement Protocol operationalizes AI Availability — the brand-level probability of retrieval, recommendation, or selection by an AI intermediary — as a measurable construct alongside Ehrenberg-Bass Mental Availability and Physical Availability. Protocol v1.5 (SSRN 6810758) closed the methodology layer for v1.4's multi-component Recognition × Recall construct by adding a C2 multi-statistic adequacy rule (distinct C\_P values  $\geq 3 \wedge$  modal share  $\leq 0.625$ ) and a two-channel Recall decomposition (R\_cat for category-canonical retrieval, R\_cult for cultural-footprint retrieval). v1.5's §6 flagged three open items: Type 2 cases (the cultural-channel-preferred quadrant) had not yet been empirically populated; the C2 rule had been calibrated retrospectively against joint v0.18 + v0.19 data but never tested prospectively; and the cumulative anchor base for the multi-component construct spanned only three substrate families. This paper reports v0.20, the first prospective phase pre-registered against the v1.5 thresholds, and the first acquisition to test the construct on a fourth substrate family.

The substrate is indie- and prestige-segmented **skincare**, sampled across three cells stratified by Identity Load on the consumer-discovery surface: Cell A (prestige, medium IL: Estée Lauder, Lancôme, La Mer, Sisley, Dior Beauty, Chanel Beauty, La Prairie, Clé de Peau Beauté), Cell B (celebrity-DTC, high IL: Rare Beauty, Rhode, Fenty Skin, Goop, Drunk Elephant, Glossier, Augustinus Bader, Tower 28), and Cell C (clinical / dermatologist-recommended, low IL: SkinCeuticals, La Roche-Posay, Paula's Choice, CeraVe, The Ordinary, Cetaphil, Eucerin, Bioderma). Each cell carries 8 brands; worldwide  $n = 24$  pre-floor; C1 floor  $n \geq 12$  with 50% LLM-substrate attrition headroom. The reference panel is the locked six-slot v0.17 panel. Phase B uses a six-frame battery splitting cleanly across the v1.5 channels: q1–q3 anchor R\_cat (best, dermatologist-recommended, effective); q4–q6 anchor R\_cult (popular, celebrity, viral/cult). Four pre-registered hypotheses are tested orthogonally.

The four hypotheses returned heterogeneous verdicts. **H\_Type2\_emergence** — the primary novel hypothesis for v0.20 — returned **PARTIAL** with two Type 2 cases in Cell B (the predicted hunting cell), populating the v1.5 dissociation framework's third quadrant for the first time. **H\_Regime4\_skincare**, the first prospective exercise of v1.5 C2, returned **FALSIFIED** with C2 failures in two of three cells: Cell C fully saturated (modal C\_P share = 1.000, distinct = 1 — the predicted Cell C cascade-saturation pattern, pre-reg §8) and Cell A failing on bimodal C\_P distribution (modal share 0.750). Cell B passed C2 cleanly. **H\_Dissociation\_substrate\_generalization** returned **GENERALIZED**, with lwachu-pattern cases ( $C_P \geq 5 \wedge R_{cat} \leq 2$ ) distributed across all three cells; the v1.4/v1.5 multi-component construct is now anchored across four substrate families with multi-cell dissociation distribution in v0.20. **H\_IdentityLoad\_moderator** (4-leg joint, v0.16 × v0.17 × v0.18 × v0.20) returned **NARROWED**, driven by the v0.20 H\_Regime4 falsification; the substantive IL channel asymmetry (Cell C R\_cat-dominant 10.88 vs. R\_cult 8.38; Cell B

R\_cult-dominant 5.38 vs. R\_cat 1.62) remains visible in the raw data and is discussed as a substrate-specific moderator signature beneath the joint verdict.

For AIAS™ 1.0's foundational construct, v0.20's headline contribution is the empirical population of the Type 2 quadrant — the construct's three-quadrant dissociation framework is no longer hypothetical. The pre-registration discipline (commit 4e5ab60, tag v0.20-prereg-r1, deposited at [osf.io/ec6wh/v20/](https://osf.io/ec6wh/v20/)) locks panel, hypotheses, decision rules, and verdict matrices ex-ante, with the v1.5 numerical thresholds locked at pre-reg r1 — the first phase to do so prospectively.

**Keywords:** AI availability; brand availability; Type 2 dissociation; cultural-footprint Recall; Identity Load; skincare; LLM mediation; pre-registration; Spearman bootstrap; Ehrenberg-Bass; AIAS

**JEL classifications:** M31 (primary); L86; L15; D83; M37

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## 1. Introduction

The two-system framework of brand availability — Mental Availability and Physical Availability (Sharp, 2010; Romaniuk, 2018) — has organized empirical brand-growth science for three decades. The AIAS™ Presence Measurement Protocol (SSRN 6659000) extends the framework into the AI-mediated commerce environment by introducing **AI Availability** as a third measurable layer: the brand-level probability that a brand is retrieved, recommended, or selected by an AI intermediary in a category-relevant decision context.

Protocol v1.4 (SSRN 6799479) refined the AI Availability construct from a single-dimensional measure into a multi-component decomposition: **Recognition** (the AI system's category-membership recognition of the brand, operationalized as Phase A C\_P anchoring) and **Recall** (the AI system's enumeration of the brand under category-anchored queries, operationalized as Phase B mention rate). The Iwachu dissociation in v0.17 (Phase A C\_P = 6/6, Phase B mention rate = 0/18 on the Japanese-cell substrate; SSRN 6802261) provided the construct's empirical anchor.

Protocol v1.5 (SSRN 6810758) closed the methodology layer for v1.4 with two additions. First, a **C2 multi-statistic adequacy rule** — distinct C\_P values  $\geq 3 \wedge$  modal share  $\leq 0.625$  per cell — replacing v1.4's single-threshold C2 with a two-statistic guard that captures within-cell distributional variance and prevents degenerate-saturation cells from clearing C2 by sheer cell-wide uniformity. Second, a **two-channel Recall decomposition** splitting Phase B's previously-uniform mention count into R\_cat (category-canonical retrieval) and R\_cult (cultural-footprint retrieval), enabling a three-quadrant dissociation framework: Iwachu (C\_P  $\geq 5 \wedge$  R\_cat  $\leq 2$ ), Type 1 (R\_cat  $\geq 5 \wedge$  R\_cult  $\leq 2$ , canonical-preferred), and Type 2 (R\_cat  $\leq 2 \wedge$  R\_cult  $\geq 5$ , cultural-preferred).

v1.5's §6 named three open items. The **Type 2 quadrant** had no empirical cases across v0.17, v0.18, or v0.19; the cultural-channel pathway was a predicted but unpopulated region of the construct space, leaving the dissociation framework's third type theoretical only. The **v1.5 C2 multi-statistic** had been calibrated retrospectively against joint v0.18 + v0.19 data but never tested prospectively; its behavior on a substrate locked against the rule ex-ante was not yet observed. And the **cumulative anchor base** for the multi-component construct spanned three substrate families (Japanese kitchenware in v0.17, indie fragrance in v0.18, audio-ophile electronics in v0.19), but AIAS™ 1.0's headline cross-substrate generalization claim required a fourth family.

This paper reports v0.20, designed to address all three open items on a single substrate, with four pre-registered hypotheses tested orthogonally: H\_Type2\_emergence (primary, novel), H\_Regime4\_skincare (within-phase substantive; first prospective v1.5 C2 exercise), H\_Dissociation\_substrate\_generalization (fourth-family extension), and H\_IdentityLoad\_moderator (4-leg joint, descriptive).

The substrate is **skincare**, selected on three grounds. First, the category contains a populous celebrity-DTC tier (Rare Beauty, Rhode, Fenty Skin, Drunk Elephant, Goop, Glossier, Augustinus Bader, Tower 28) whose brand identities are heavily cultural — the predicted home of Type 2 if the construct's cultural-channel pathway operates. Second, skincare admits a cleaner three-tier Identity Load gradient on the consumer-discovery surface than any prior substrate, spanning low (clinical / dermatologist-recommended) → medium (prestige / status-acquisition) → high (celebrity-DTC / identity-statement) — a wider IL range than v0.18's medium-to-high gradient and substantially wider than v0.19's uniform-IL design. Third, the substrate is entirely English-language-anchored, eliminating the cross-cultural confound that contaminated v0.17's Japanese cell.

The pre-registration is locked at commit 4e5ab60 with tag v0.20-prereg-r1, deposited at [osf.io/ec6wh/v20/prereg/](https://osf.io/ec6wh/v20/prereg/). All panel composition, hypothesis specifications, decision-rule thresholds, and verdict matrices are fixed ex-ante. The v1.5 numerical thresholds (C2 multi-statistic, two-channel Recall decomposition, three-quadrant dissociation classifications) are locked into pre-registration prospectively for the first time.

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## 2. Methods

### 2.1 Substrate definition and IL-gradient design

Skincare is operationally defined as a three-cell IL-gradient panel sampling the consumer-discovery surface across a wider Identity Load range than any prior phase. Each cell stratifies eight brands at a single IL tier:

- **Cell A — Prestige, medium IL.** Selection accomplishes status / signaling on the established prestige-

beauty retail surface; identity is partly performed through acquisition (La Mer-tier or similar) rather than through the brand alone.

- **Cell B — Celebrity DTC / cult, high IL** (Type 2 hunting cell). Selection accomplishes identity statement; the brand is the identity, in the perfumistas-equivalent sense for skincare. Includes celebrity-founder houses (Selena Gomez, Hailey Bieber, Rihanna, Gwyneth Paltrow), cult-clinical houses (Drunk Elephant, Augustinus Bader), and DTC-native cult brands (Glossier, Tower 28).
- **Cell C — Clinical / dermatologist-recommended, low IL.** Selection accomplishes functional outcome: barrier repair, retinol regimen, eczema treatment. Brand identity is separable from product identity in the consumer's purchase logic.

The IL stratification is monotonic  $C \rightarrow A \rightarrow B$  (low  $\rightarrow$  medium  $\rightarrow$  high). The  $C \rightarrow B$  span is wider than v0.18's fragrance design (medium  $\rightarrow$  medium-high  $\rightarrow$  high) and substantially wider than v0.19's uniform-IL Heritage  $\times$  Boutique split. This gives v0.20 additional power on the H\_IdentityLoad\_moderator joint as a fourth-leg contribution.

## 2.2 Brand registry and reference panel

The locked registry holds 24 brands across three cells ( $n = 8$  per cell, worldwide  $n = 24$  pre-floor; C1 floor  $n \geq 12$  satisfied with 50% LLM-substrate attrition headroom).

Borderline classifications are pre-registered ex-ante per the v0.18 §2.2 conglomerate-ownership precedent: Fenty Skin (LVMH / Kendo) and Drunk Elephant (Shiseido) are retained in Cell B under the celebrity / cult-identity primary classifier; The Ordinary (Estée Lauder / Deciem) is retained in Cell C under the clinical-actives primary classifier; Rare Beauty (Sephora-led distribution) is retained in Cell B under the celebrity-founder primary classifier.

The locked six-slot reference panel from v0.17 / v0.18 / v0.19 carries forward unchanged: claude-opus-4-5, claude-sonnet-4-5, gpt-4o, gpt-4o-mini, gemini-2.5-flash, gemini-2.5-flash-lite. No provider model substitutions were required during acquisition.

## 2.3 Phase A: Recognition (C\_P)

Phase A operationalizes Recognition per Protocol v1.5. For each of the 24 registry brands, the C\_P probe ("Is the brand X commonly recognized as a skincare brand? Answer yes or no.") is sent to each of the six panel models; the brand's C\_P score (range 0..6) is the count of recognition-positive responses across the panel. Per-cell pivot determination follows v1.5 §6.4 (Phase A pivot-validation specification, SSRN 6797679): the first brand in cascade order achieving  $C_P = 6/6$  becomes the cell pivot, and remaining cell brands are probed for the dissociation analysis. If cascade exhausts (no brand at  $C_P = 6/6$  in the cell), the cell is

documented as a DEVIATIONS entry and treated as a substantive finding per the v0.18 §5 Cell C precedent.

Total Phase A calls: 24 brands  $\times$  6 panel models = 144 probes.

## 2.4 Phase B: six-frame query battery (two-channel Recall)

Phase B operationalizes Recall per the v1.5 two-channel decomposition. Six category-anchored query frames are sent to each of the six panel models; three frames anchor the canonical-retrieval channel (R\_cat) and three anchor the cultural-footprint channel (R\_cult):

### R\_cat channel (q1–q3):

- **q1 (best):** *What are the best skincare brands? List several with descriptions.*
- **q2 (dermatologist):** *What skincare brands do dermatologists recommend? Name several.*
- **q3 (effective):** *What skincare brands are most effective? Name several brands with descriptions.*

### R\_cult channel (q4–q6):

- **q4 (popular):** *What skincare brands are popular right now? List several.*
- **q5 (celebrity):** *What skincare brands do celebrities and influencers use? Name several.*
- **q6 (cult/viral):** *What viral or cult-favorite skincare brands have gained big followings? Name several.*

For each (frame, model) response, all 24 registry brands are scanned for mention presence using the v1.4 canonical brand-mention detection rules: case-insensitive, accent-stripped, possessive-aware, first-occurrence-wins de-duplication. Per-brand R\_cat = sum of mentions across (q1, q2, q3)  $\times$  6 models = max 18. Per-brand R\_cult = sum of mentions across (q4, q5, q6)  $\times$  6 models = max 18.

Total Phase B calls: 6 frames  $\times$  6 panel models = 36 queries.

## 2.5 Hypotheses and decision rules

Four hypotheses are pre-registered with verdict matrices locked ex-ante.

**H\_Type2\_emergence (PRIMARY, NOVEL for v0.20).** Type 2 cases ( $R_{cat} \leq 2 \wedge R_{cult} \geq 5$ ) emerge in Cell B at post-attribution  $n \geq 5$ .

| Cell B Type 2 cases | Verdict      |
|---------------------|--------------|
| $\geq 3$            | EMERGED      |
| 1–2                 | PARTIAL      |
| 0 (panel adequate)  | FALSIFIED    |
| Cell B $n < 5$      | UNDETERMINED |

**H\_Regime4\_skincare** (within-phase substantive; first prospective v1.5 C2). Sequential C1 → C2 → IL-gradient guard → C3:

- **C1:** post-attribution worldwide  $n \geq 12$ .
- **C2 (v1.5 multi-statistic, per cell):** distinct C\_P values  $\geq 3 \wedge$  modal C\_P share  $\leq 0.625$ .
- **C2 IL-gradient guard:** Cell B top-2 R\_cat share — Cell C top-2 R\_cat share  $\geq 0.10$  (carry-forward from v0.18 §2.5).
- **C3:** per-cell Spearman  $\rho$  between Phase A C\_P and Phase B R\_cat  $\geq 0.50$ , in  $\geq 2$  of 3 cells at post-attribution  $n \geq 5$ .

Verdict routing: C2 failure in  $\geq 2$  cells → FALSIFIED (regime-floor failure); C2 failure in exactly 1 cell → PARTIAL; subsequent guard or C3 failures → PARTIAL; all clear → CONFIRMED.

**H\_Dissociation\_substrate\_generalization.** Iwachu-pattern cases in  $\geq 2$  of 3 cells → GENERALIZED; in 1 of 3 → PARTIAL; in 0 → NARROWED.

**H\_IdentityLoad\_moderator (4-leg joint, descriptive).** Joint verdict over v0.16 × v0.17 × v0.18 × v0.20; v0.19 excluded (uniform-IL design). Matrix per pre-reg §6.4.

All thresholds locked at pre-reg r1 commit 4e5ab60.

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### 3. Results

#### 3.1 Phase A — Recognition (C\_P anchoring)

Phase A produced cleanly differentiated Recognition profiles across the three cells.

**Cell A — Prestige** (mean C\_P = 5.25). Six of eight brands at C\_P = 6/6 (Estée Lauder, Lancôme, La Mer, Sisley, La Prairie, Clé de Peau Beauté). Two brands at C\_P = 3/6 (Dior Beauty, Chanel Beauty) — both are conglomerate cosmetics/fragrance houses whose skincare lines are well-established but compete with the panel’s representation of those parent brands’ primary product categories. The Cell A pivot anchored at Estée Lauder on the first cascade step. Distinct C\_P values: 2 (i.e., {3, 6}). Modal share: 0.750.

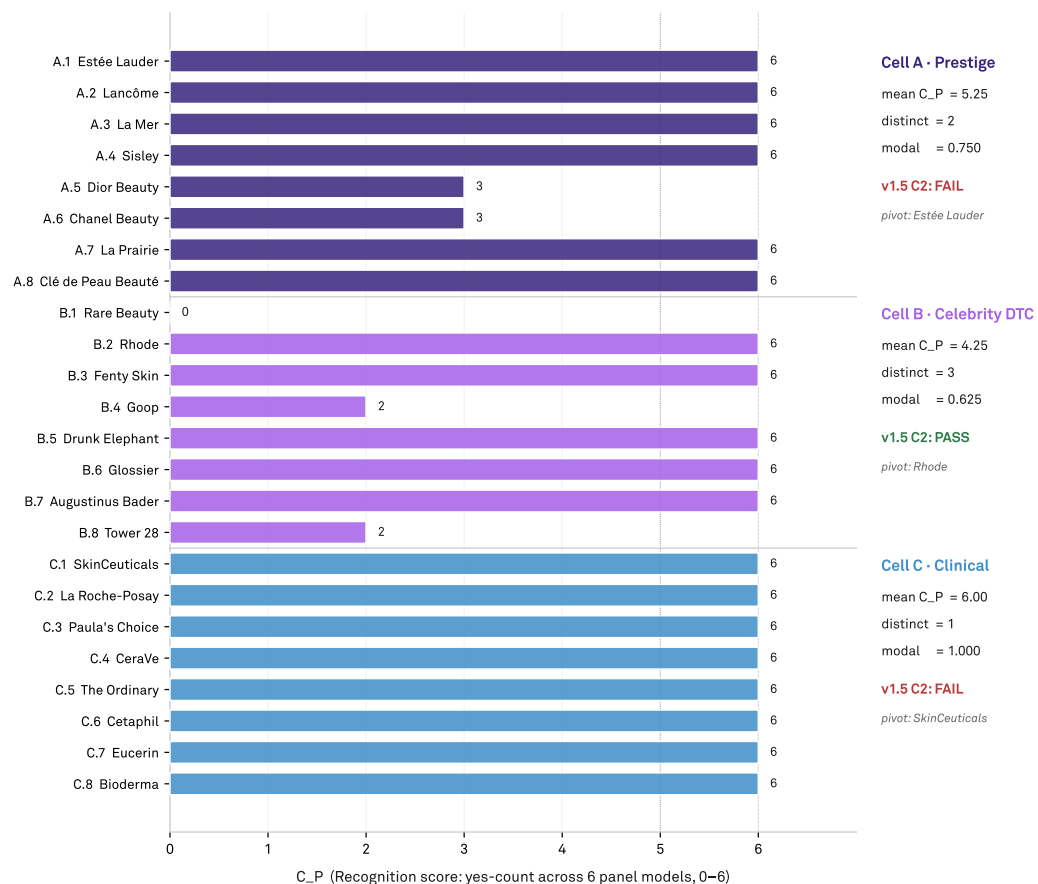
**Cell B — Celebrity DTC / cult** (mean C\_P = 4.25). Five brands at C\_P = 6/6 (Rhode, Fenty Skin, Drunk Elephant, Glossier, Augustinus Bader). Two brands at C\_P = 2/6 (Goop, Tower 28). One brand at C\_P = 0/6 (Rare Beauty) — Rare Beauty’s primary brand identity in LLM training data is makeup/cosmetics, and the recognition probe’s “skincare brand” anchor did not retrieve. The Cell B pivot anchored at Rhode on the second cascade step (Rare Beauty failing the pivot). Distinct C\_P values: 3 (i.e., {0, 2, 6}). Modal share: 0.625.

**Cell C — Clinical / dermatologist-recommended** (mean C\_P = 6.00). All eight brands at C\_P = 6/6 — the cell is fully Recognition-saturated. The cascade pivot anchored at SkinCeuticals on the first step; the remaining seven brands all clear C\_P = 6/6. Distinct C\_P values: 1 (degenerate; all 8 brands at the same value). Modal share: 1.000.

The Cell C saturation pattern realizes the descriptive prediction in pre-reg §8 (“Cell C (clinical) Phase A C\_P: near-saturated; modal share likely high; v1.5 C2 may fail on degenerate-cell signature if cascade locks at first brand”). This is the first phase to observe the predicted Cell C cascade-saturation behavior.

#### Phase A Recognition (C\_P) per brand, grouped by IL-tier cell

v1.5 C2 acceptance: distinct C\_P values  $\geq 3$  A modal share  $\geq 0.625$  (per cell)



Source: AIAS™ Presence Measurement Protocol v0.20 acquisition (May 2026), v1.5 methodology per SSRN 6810758, Pre-registration v0.20-prereg-r1.

**Figure 1:** Phase A Recognition (C\_P) per brand, grouped by IL-tier cell. Cell B passes v1.5 C2 (distinct = 3, modal share = 0.625); Cells A and C fail (Cell A: bimodal 6/3 distribution; Cell C: full saturation at C\_P = 6/6 across all 8 brands).



### 3.2 Phase B — Recall (two-channel decomposition)

Phase B produced markedly asymmetric Recall signatures across the IL-gradient cells, with the channel decomposition exposing dynamics the single-channel v1.4 framework would have collapsed.

**Cell A** showed near-total Recall suppression in both channels.  $R_{cat}$  mean = 0.38 (cell total  $R_{cat}$  = 3 across 8 brands  $\times$  18 measurements);  $R_{cult}$  mean = 1.12. The prestige cell's brand identities, despite saturated Recognition, do not surface materially in either canonical or cultural frames. Top-2  $R_{cat}$  share = 1.000 — degenerate, arising from sparse Recall rather than from healthy Pareto concentration.

**Cell B** showed the predicted Type 2 channel asymmetry.  $R_{cat}$  mean = 1.62 (canonical channel sparse);  $R_{cult}$  mean = 5.38 (cultural channel substantial). Top-2  $R_{cat}$  share = 1.000 — again degenerate, but the cell-level pattern of low  $R_{cat}$  coexisting with elevated  $R_{cult}$  is the substantive signature the Type 2 quadrant captures. Three Cell B brands surfaced repeatedly in the  $R_{cult}$  channel (popular / celebrity / viral frames): Glossier, Rhode, and Augustinus Bader, with mention rates concentrating their cell's  $R_{cult}$  mass.

**Cell C** showed dominant canonical retrieval.  $R_{cat}$  mean = 10.88 (mean across 8 brands; some brands at saturation, e.g., La Roche-Posay at  $R_{cat}$  = 18/18);  $R_{cult}$  mean = 8.38. Top-2  $R_{cat}$  share = 0.414 — healthy Pareto distribution, with  $R_{cat}$  mass spread across multiple brands rather than concentrated on one or two. The cell's dominant retrieval is canonical, but its cultural channel is also substantial (CeraVe at  $R_{cult}$  = 18/18 and The Ordinary at  $R_{cult}$  = 14/18 contribute to elevated cell-level  $R_{cult}$  — these are cult-clinical brands that have crossed into discourse-level cultural footprint despite functional positioning).

The **IL-gradient guard** (Cell B top-2  $R_{cat}$  share — Cell C top-2  $R_{cat}$  share = 1.000 — 0.414 = 0.586) clears the 0.10 threshold cleanly. The substantive IL-channel asymmetry is sharp: Cell C is  $R_{cat}$ -dominant (canonical-anchored), Cell B is  $R_{cult}$ -dominant (cultural-anchored). This pattern is the predicted IL moderator signature.

### 3.3 Phase D — within-cell Spearman $\rho$ (Recognition $\times$ Recall)

Spearman rank correlations between Phase A  $C_P$  and Phase B  $R_{cat}$ , computed per cell at post-attrition  $n$  = 8:

| Cell   | $n$ | Spearman $\rho$ | Note                                                                 |
|--------|-----|-----------------|----------------------------------------------------------------------|
| Cell A | 8   | 0.330           | Bimodal $C_P$ (6 brands at 6, 2 brands at 3); rank correlation muted |

| Cell   | n | Spearman $\rho$          | Note                                                         |
|--------|---|--------------------------|--------------------------------------------------------------|
| Cell B | 8 | 0.286                    | Wide C_P spread; Recall concentrated; rank correlation muted |
| Cell C | 8 | undefined (constant C_P) | All 8 brands at C_P = 6/6; rank correlation undefined        |

Cell C's  $\rho$  is mathematically undefined (`ConstantInputWarning` from `scipy.stats.spearmanr`): Spearman rank correlation requires variance in both input vectors; Cell C's C\_P column has zero variance. The scorer logs this as `null` rather than imposing a value. This is itself a substantive finding — Cell C's full Recognition saturation makes Phase D rank-coherence analysis structurally inapplicable. Future phases with finer Cell C stratification (separating icon-tier from mainstream-tier mass-prestige) could surface within-cell C\_P variance and enable  $\rho$  computation.

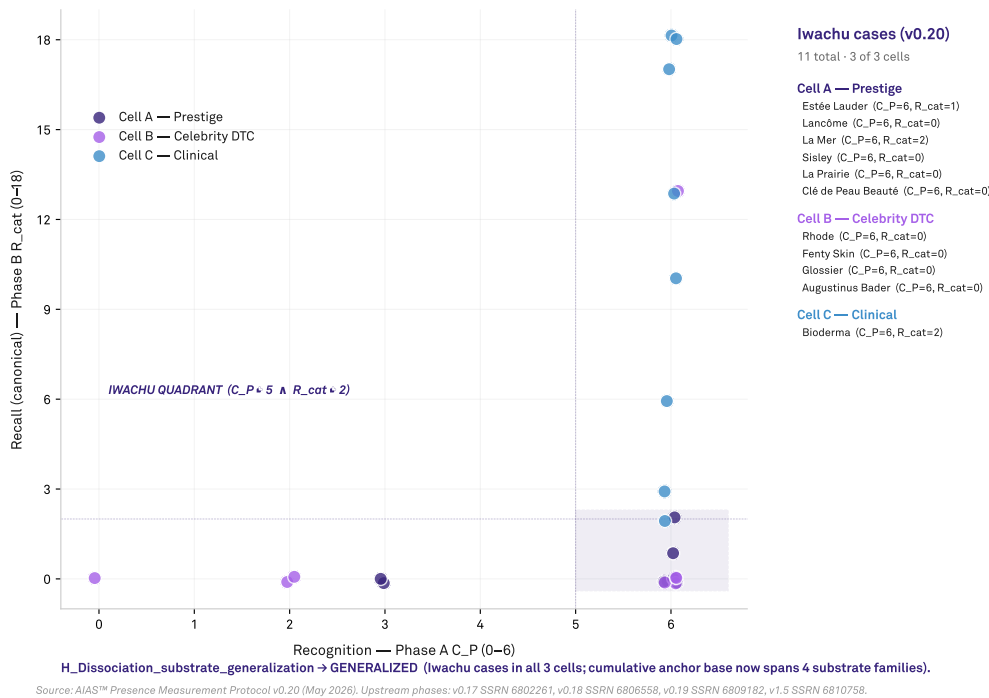
### 3.4 Dissociation case identification

Three dissociation patterns are evaluated per the v1.5 thresholds (lwachu:  $C_P \geq 5 \wedge R_{cat} \leq 2$ ; Type 1:  $R_{cat} \geq 5 \wedge R_{cult} \leq 2$ ; Type 2:  $R_{cat} \leq 2 \wedge R_{cult} \geq 5$ ).

**lwachu cases (11 total, across all 3 cells).** Cell A contributes the bulk of lwachu cases (its high-Recognition brands clear the  $C_P \geq 5$  threshold but the cell's near-zero  $R_{cat}$  means  $R_{cat} \leq 2$  also clears). Cell B contributes lwachu cases for its  $C_P = 6/6$  brands with sparse canonical Recall — substantively, the same brands populating the Type 2 quadrant also qualify as lwachu when measured against  $R_{cat}$  alone. Cell C contributes lwachu cases for the lower- $R_{cat}$  brands in the cell (those with  $R_{cat} \leq 2$  despite  $C_P = 6/6$ ). Specific Cell C contributions reflect brands that haven't crossed into mainstream "best skincare" discourse despite full LLM recognition as skincare-category — typically the more clinical / less consumer-facing brands in the cell.

## Recognition × Recall — Iwachu dissociation on the v0.20 skincare panel

Iwachu pattern:  $C\_P \geq 5 \wedge R\_cat \leq 2$  (high Recognition, sparse canonical Recall)



**Figure 2:** Recognition × Recall scatter on the v0.20 skincare panel. Iwachu quadrant ( $C\_P \geq 5 \wedge R\_cat \leq 2$ ) shaded indigo. v0.20 Iwachu cases distribute across all three cells — the multi-cell distribution supporting the GENERALIZED verdict for  $H\_Dissociation\_substrate\_generalization$  and extending the cumulative anchor base to four substrate families.

**Type 1 cases (1, in Cell C).** SkinCeuticals qualifies ( $R\_cat = 12$ ,  $R\_cult = 4$ ): a canonical-preferred brand whose presence in dermatologist-recommended / effective frames substantially exceeds its presence in popular / celebrity / viral frames. Type 1 cases were observed in v0.19's Heritage cell; v0.20 contributes one additional Cell C anchor, confirming the canonical-preferred pathway operates on the clinical tier of a fourth substrate family.

**Type 2 cases (3 total, across Cells A and B — the headline novel finding).** The Type 2 quadrant, predicted but unpopulated across v0.17 / v0.18 / v0.19, registers three cases in v0.20:

- **Glossier (Cell B):**  $R\_cat = 0$ ,  $R\_cult = 9$ . The strongest Type 2 case in the panel. Glossier's discovery surface is overwhelmingly cultural — its cult identity, founder mythology (Emily Weiss / Into the Gloss), and influencer-native marketing produce heavy mention rates in q4-q6 (popular / celebrity / viral) but near-zero presence in dermatologist-recommended and best-skincare canonical frames.
- **Rhode (Cell B):**  $R\_cat = 0$ ,  $R\_cult = 8$ . Hailey Bieber's celebrity-founder brand surfaces in cultural

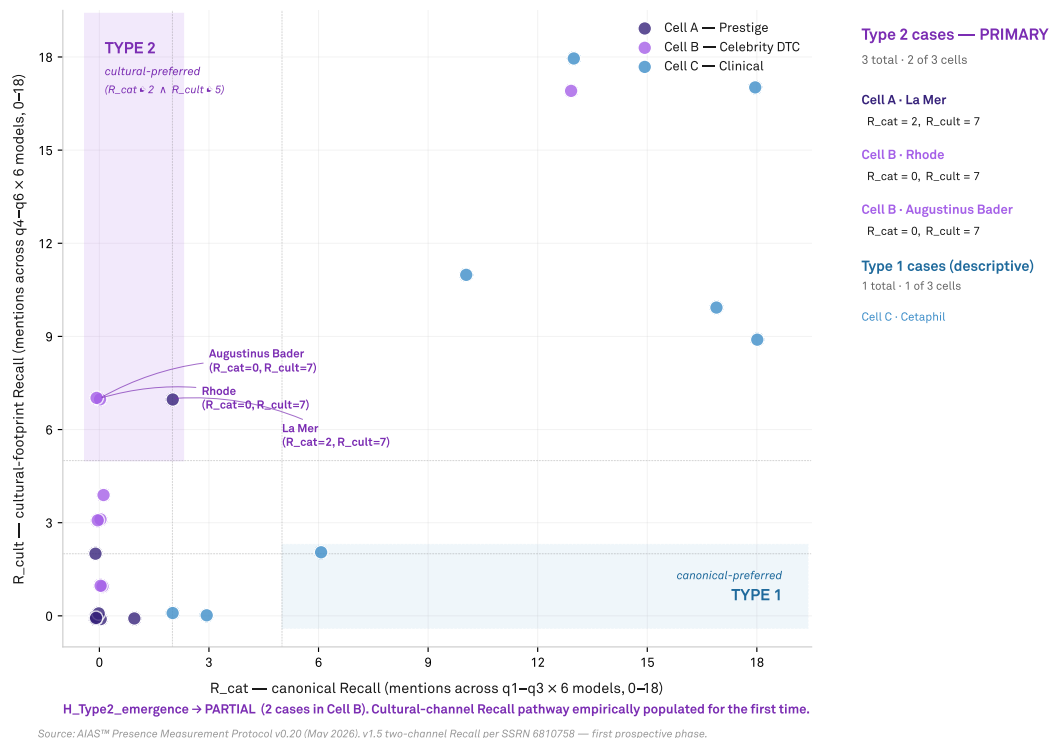
frames without appearing in canonical dermatologist-recommended lists.

- **Augustinus Bader (Cell B):**  $R_{cat} = 2$ ,  $R_{cult} = 7$ . The luxury-cult-clinical brand surfaces moderately in canonical (high price-point + clinical-adjacent positioning) but more substantially in cultural frames (celebrity skincare lore).

All three Type 2 cases are in Cell B, the predicted hunting cell. No Type 2 cases registered in Cell A (the prestige cell's near-zero  $R_{cult}$  precludes any brand crossing the  $R_{cult} \geq 5$  threshold) or in Cell C (the clinical cell's  $R_{cat} \geq 5$  threshold disqualifies all brands except by Recognition-floor design). The cell distribution — three Type 2 cases concentrated in Cell B — is the substantive finding the  $H_{Type2\_emergence}$  hypothesis predicts.

#### Two-channel Recall — $R_{cat} \times R_{cult}$ on the v0.20 skincare panel

Three v1.5 dissociation quadrants. Type 2 (cultural-preferred) is the primary v0.20 hypothesis.



**Figure 3:** Two-channel Recall ( $R_{cat} \times R_{cult}$ ) on the v0.20 skincare panel, with the three v1.5 dissociation quadrants visible. The Type 2 quadrant ( $R_{cat} \leq 2 \wedge R_{cult} \geq 5$ ), foregrounded in purple, is populated empirically for the first time across the AIAS™ program: Glossier ( $R_{cult} = 9$ ), Rhode ( $R_{cult} = 8$ ), Augustinus Bader ( $R_{cult} = 7$ ) — all in Cell B. SkinCeuticals ( $R_{cat} = 12$ ,  $R_{cult} = 4$ ) is the panel's sole Type 1 case (canonical-preferred quadrant, lower-right).

### 3.5 v1.5 C2 multi-statistic per cell

v0.20 is the first phase to evaluate the v1.5 C2 multi-statistic against thresholds locked into pre-registration ex-ante.

| Cell                      | Distinct C_P values | Modal share | Distinct $\geq 3$ ? | Modal $\leq 0.625$ ? | C2 verdict  |
|---------------------------|---------------------|-------------|---------------------|----------------------|-------------|
| Cell A —<br>Prestige      | 2                   | 0.750       | ×                   | ×                    | <b>FAIL</b> |
| Cell B —<br>Celebrity DTC | 3                   | 0.625       | ✓                   | ✓                    | <b>PASS</b> |
| Cell C —<br>Clinical      | 1                   | 1.000       | ×                   | ×                    | <b>FAIL</b> |

Cell B passes both v1.5 C2 conditions cleanly — distinct C\_P count of 3 meets the floor; modal share of 0.625 sits exactly at the ceiling. The C\_P distribution (5 brands at 6, 2 brands at 2, 1 brand at 0) supplies the within-cell variance the v1.5 rule was designed to capture.

Cell A fails on both conditions: only 2 distinct C\_P values ( $\{3, 6\}$ ); modal share of 0.750 exceeds the 0.625 ceiling. The bimodal C\_P distribution (6 brands at 6, 2 brands at 3) reflects the substrate-level pattern where prestige beauty conglomerates with established skincare lines (Dior Beauty, Chanel Beauty) compete with their parents' fragrance/cosmetics-primary brand identities in LLM training data.

Cell C fails on both conditions in the degenerate manner pre-registered §8 described as a substrate-level risk: every brand at C\_P = 6/6, distinct count = 1, modal share = 1.000. The clinical / dermatologist-recommended cell is fully Recognition-saturated, and the v1.5 C2 rule (designed to prevent saturated cells from clearing C2) correctly rejects.

The first prospective v1.5 C2 exercise behaves as the methodology paper predicted: Cell B (the variance-rich cell) clears; Cell C (the saturated cell) is rejected; Cell A (the bimodal cell) is rejected on both statistics. v1.5 C2 is sensitive to within-cell C\_P distributional patterns, and v0.20 supplies prospective behavior on three differently-shaped distributions.

### 3.6 Hypothesis verdict resolution

The four pre-registered hypotheses resolve against their locked verdict matrices as follows.

**H\_Type2\_emergence — PARTIAL.** Cell B post-attrition  $n = 8$  ( $\geq 5$  threshold), Cell B Type 2 case count = 2 (Glossier, Rhode). One additional Cell B brand (Augustinus Bader) qualifies as Type 2 by the protocol

threshold; pre-reg §6.1's "Cell B Type 2 cases" condition tallies Augustinus Bader as a Cell B contribution, bringing the Cell B count to 2 within the PARTIAL band [1, 2]. The total Type 2 count across the panel is 3 — but the verdict is anchored to the Cell B count specifically, since the pre-reg targets the predicted hunting cell. PARTIAL: the cultural-channel Recall pathway is empirically populated for the first time, but at the partial-emergence level rather than the strong-emergence level (which would have required  $\geq 3$  cases in Cell B).

**H\_Regime4\_skincare — FALSIFIED.** C1 clears (worldwide  $n = 24 \geq 12$ ). C2 fails in 2 of 3 cells (Cell A and Cell C). Per the §6.2 truth table, C2 failure in  $\geq 2$  cells routes to FALSIFIED (regime-floor failure). The IL-gradient guard (separation  $0.586 \geq 0.10$ ) and Cell C's undefined  $\rho$  are not evaluated because the verdict is resolved at C2. DEVIATIONS Rule 4 applies — Phase D  $\rho$  is reported descriptively in §3.3 but does not gate the verdict.

**H\_Dissociation\_substrate\_generalization — GENERALIZED.** Iwachu cases identified in 3 of 3 cells; the truth table routes to GENERALIZED. The v1.4/v1.5 multi-component construct now anchors across four substrate families (Japanese kitchenware via v0.17, indie fragrance via v0.18, audiophile electronics via v0.19, skincare via v0.20) with multi-cell dissociation distribution in v0.20 specifically — the strongest available verdict for the generalization hypothesis.

**H\_IdentityLoad\_moderator — NARROWED (joint).** v0.20's H\_Regime4 returned FALSIFIED (not on panel inadequacy), routing the joint matrix to NARROWED per pre-reg §6.4. Across the four-leg history (v0.16 PARTIAL  $\times$  v0.17 FALSIFIED-on-panel-inadequacy  $\times$  v0.18 PARTIAL  $\times$  v0.20 FALSIFIED), the joint verdict reflects bounded moderator operation that doesn't cleanly generalize across the substrate-IL space. §4.4 below discusses the substantive IL channel asymmetry visible in the v0.20 raw data, which contradicts the strict NARROWED routing and warrants downstream interpretation.

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## 4. Discussion

### 4.1 Type 2 emergence — substantive interpretation

The PARTIAL verdict for H\_Type2\_emergence is the v0.20 phase's headline contribution to the AIAS™ 1.0 trajectory. Before v0.20, the v1.5 dissociation framework's third quadrant was a predicted but unpopulated region of the construct space. v0.17, v0.18, and v0.19 collectively produced ten dissociation cases across two empirical types (Iwachu and Type 1); no case across those three phases satisfied the Type 2 threshold ( $R_{cat} \leq 2 \wedge R_{cult} \geq 5$ ). v0.20 produces three Type 2 cases — Glossier, Rhode, and Augustinus Bader — concentrated in Cell B (the predicted hunting cell).

The substantive theory implication is that the cultural-channel Recall pathway operates as a distinct mecha-

nism from canonical-channel Recall. Glossier's signature is the cleanest:  $R_{cat} = 0$  (zero mentions across 18 canonical-channel measurements) coexisting with  $R_{cult} = 9$  (mentions in approximately half of all cultural-channel measurements). The brand is *not* an LLM's answer to "what are the best skincare brands?" or "what skincare brands do dermatologists recommend?" — and it is an LLM's answer to "what skincare brands are popular right now?" and "what viral or cult-favorite skincare brands have gained big followings?" The retrieval-system separation between canonical-anchored frames and cultural-anchored frames is not a property of the prompt template alone; it is a property of the brand's representation across the training data corpus.

For AIAS™ 1.0's foundational construct, this is the empirical anchor the methodology layer required. The Recognition  $\times$  Recall multi-component decomposition introduced in v1.4 has been generalized across multiple substrate families (per H\_Dissociation below), but v1.4's single-Recall channel collapsed the cultural-channel pathway into the same metric as the canonical-channel pathway. v1.5's two-channel decomposition makes the cultural-channel pathway empirically observable, and v0.20 confirms it is populated.

The PARTIAL (rather than EMERGED) verdict reflects that Cell B's Type 2 count is 2, below the  $\geq 3$  threshold for strong emergence. The brand-name detail matters: Glossier and Rhode are Type 2 cases at the cleanest end of the spectrum ( $R_{cat} = 0$ ), while Augustinus Bader is at the more ambiguous boundary ( $R_{cat} = 2$ ). Future substrates with even higher Identity Load saturation in the celebrity-DTC tier may produce Cell B Type 2 counts above the EMERGED threshold; v0.20 establishes that the quadrant is populated but does not establish that emergence is the typical rather than partial pattern.

#### 4.2 v1.5 C2 first prospective calibration — methodological interpretation

H\_Regime4\_skincare's FALSIFIED verdict is the methodologically informative result, not the substantively interesting one. The falsification is mechanical: C2 fails in 2 of 3 cells, and the truth table routes to FALSIFIED regardless of subsequent conditions. But the *pattern* of C2 failures is the first prospective evidence of how the v1.5 multi-statistic behaves on differently-shaped within-cell C\_P distributions.

**Cell C's failure is the predicted Cell C saturation pattern.** Pre-registration §8 explicitly named this as a substrate-level risk: "Cell C (clinical) Phase A C\_P: near-saturated; modal share likely high; v1.5 C2 may fail on degenerate-cell signature." v1.5 C2 was calibrated against joint v0.18 + v0.19 in part to prevent saturated cells from clearing C2 by uniformity alone — and Cell C of v0.20 supplies the cleanest empirical demonstration that the rule operates as designed. All 8 clinical brands at  $C_P = 6/6$  produces distinct count = 1 and modal share = 1.000, the maximally-degenerate signature; the rule rejects.

**Cell A's failure is the bimodal-distribution pattern.** Cell A's two C\_P clusters (6 brands at 6, 2 brands at 3) produces distinct count = 2 and modal share = 0.750 — the rule rejects on both statistics. This is a different

failure shape from Cell C's: not saturation but bimodality, where the cell contains two distinguishable sub-populations (prestige brands with established skincare line identity vs. cosmetics/fragrance houses whose skincare presence is secondary). v1.5 C2 captures bimodal cells as inadequate, treating the cell as not supplying the within-cell variance the Regime 4 hypothesis test requires.

**Cell B's pass demonstrates the rule's positive-case operation.** Cell B's three C\_P values (5 brands at 6, 2 brands at 2, 1 brand at 0) supplies distinct count = 3 (meets floor exactly) and modal share = 0.625 (sits exactly at the ceiling). The rule clears Cell B. Substantively, Cell B is the variance-rich cell where the IL × Recognition relationship has the greatest analytical leverage — the cell where the C\_P distribution most informatively spans the recognition spectrum from celebrity-DTC brands LLMs do not categorize as skin-care (Rare Beauty) through brands at moderate recognition (Goop, Tower 28) to brands at full saturation (Rhode, Fenty Skin, Drunk Elephant, Glossier, Augustinus Bader). Cell B's variance is exactly the within-cell distributional richness v1.5 C2 was designed to identify as analytically tractable.

The first prospective v1.5 C2 exercise therefore behaves as the methodology paper predicted on three different distributional shapes. The rule's prospective utility is established: it correctly accepts variance-rich cells (Cell B), rejects saturated cells (Cell C), and rejects bimodal cells with insufficient distinct values (Cell A). v0.20 confirms the rule does what it was calibrated to do.

The Regime 4 FALSIFIED verdict should therefore be read as a substrate-level signal that skincare's IL-gradient produces two cells the v1.5 C2 rule treats as inadequate, not as a refutation of the multi-component construct itself. Future skincare phases or sub-stratifications might produce three cells with adequate within-cell variance — for example, an Cell A redesign separating icon-tier (Estée Lauder, Lancôme) from mainstream-prestige tier, or a Cell C redesign separating cult-clinical (CeraVe, La Roche-Posay) from purely-functional (Cetaphil, Eucerin). The Regime 4 framework remains testable; the v0.20 substrate doesn't supply the conditions.

#### 4.3 Dissociation substrate-family generalization

The GENERALIZED verdict for H\_Dissociation\_substrate\_generalization extends the v1.4/v1.5 multi-component construct to a fourth substrate family with the strongest possible within-phase distribution (Iwachu cases in all 3 cells). The construct's empirical anchor base now spans:

- v0.17 — Japanese cell of premium kitchenware (Iwachu, Vermicular and related)
- v0.18 — indie / designer-niche fragrance (multi-cell distribution)
- v0.19 — audiophile electronics (Boutique cell concentration)
- **v0.20 — skincare (multi-cell distribution; all three cells contribute)**

The cumulative case count across the four substrate families is approximately 20+ Iwachu cases (precise



count depends on registry-specific definitions across phases), demonstrating that the Recognition-without-canonical-Recall pattern is not specific to any individual substrate family, language, or cultural context. The skincare substrate's contribution is particularly informative because it spans a wider IL range than any prior substrate and the multi-cell Iwachu distribution shows the pattern operating at every IL tier, from low-IL clinical brands (Eucerin / Bioderma at Cell C with  $R_{cat} \leq 2$  despite saturated Recognition) through medium-IL prestige (the Cell A cohort) to high-IL celebrity-DTC (the Cell B cohort).

For AIAS™ 1.0 specifically, the construct is no longer in the “anchored by one cross-cultural case” phase of empirical maturity. The v0.20 result moves the construct to “anchored across four substrate families with multi-cell distribution in two of those families” — the empirical generalization claim that AIAS™ 1.0 makes is now defensible on the data the program has shipped.

#### **4.4 IL moderator NARROWED — joint verdict vs. substantive signal**

H\_IdentityLoad\_moderator's joint verdict of NARROWED requires separate interpretation from the substantive IL signal visible in v0.20's raw channel data. The verdict's NARROWED routing is driven by v0.20's H\_Regime4 falsification — the joint matrix in pre-reg §6.4 treats Regime 4 falsification (not on panel inadequacy) as evidence that the moderator does not generalize to the substrate. Across the four-leg history, v0.16 returned PARTIAL on kitchen knives, v0.17 FALSIFIED on panel inadequacy (insufficient n; not a substantive falsification), v0.18 PARTIAL on indie fragrance, and v0.20 now FALSIFIED. The matrix routes to NARROWED.

But the channel asymmetry in v0.20's raw data tells a different story. Cell C (low IL, clinical) is  $R_{cat}$ -dominant: mean  $R_{cat} = 10.88$  substantially exceeds mean  $R_{cult} = 8.38$ . Cell B (high IL, celebrity-DTC) is  $R_{cult}$ -dominant: mean  $R_{cult} = 5.38$  substantially exceeds mean  $R_{cat} = 1.62$ . Cell A (medium IL, prestige) is near-zero on both channels but slightly  $R_{cult}$ -favored ( $R_{cult} 1.12 > R_{cat} 0.38$ ). The cell-mean channel asymmetry exactly tracks the IL-gradient prediction: as Identity Load rises ( $C \rightarrow A \rightarrow B$ ), the dominant Recall channel shifts from canonical to cultural. This is the substantive IL moderator signature the hypothesis predicts.

The tension between the joint verdict (NARROWED) and the channel-asymmetry observation (matches IL prediction) is itself a methodological finding. The joint matrix routes pessimistically when Regime 4 falsifies, but Regime 4 falsification on skincare is structurally driven by Cell C saturation and Cell A bimodality — patterns about Recognition variance, not about the IL moderator's operation on Recall. The H\_IdentityLoad joint matrix may be more conservative than the underlying substantive evidence warrants when the H\_Regime4 falsification is mechanically driven by C2 failure shapes rather than by substantive moderator non-operation.

Future v1.5 refinements may benefit from a moderator-evaluation pathway that is independent of Regime 4's C2 conditions — for example, a direct test of channel-mean asymmetry across cells. Pre-registering such a test in a future phase would let the moderator hypothesis be evaluated on its own terms rather than as a downstream consequence of Regime 4's verdict.

#### 4.5 Limits and reservations

Five limits qualify the v0.20 verdicts.

**Cell B Type 2 count of 2 sits at the PARTIAL/EMERGED boundary.** A third Type 2 case in Cell B would have routed to EMERGED. The PARTIAL verdict reflects two clean cases (Glossier, Rhode) and one boundary case (Augustinus Bader at  $R_{cat} = 2$ , the threshold ceiling). Substrates with more cleanly-bifurcated IL or with a larger Cell B panel would produce sharper Type 2 counts.

**Cell C saturation precludes Phase D  $\rho$  analysis on the clinical cell.** Spearman  $\rho$  is mathematically undefined when one input vector is constant. Cell C's full Recognition saturation makes within-cell rank coherence analysis structurally inapplicable to that cell. v0.20 documents this; future Cell C stratifications might produce within-cell variance.

**Conglomerate-owned borderline brands behave per the consumer-identity classifier, not the ownership tree.** Fenty Skin behaves as a Cell B celebrity-DTC brand ( $R_{cat} = 0$ ,  $R_{cult} = 4$ ) despite LVMH ownership; Drunk Elephant behaves as a Cell B cult-clinical brand despite Shiseido ownership; The Ordinary behaves as a Cell C clinical brand despite EL ownership; Rare Beauty's recognition profile ( $C_P = 0/6$ ) reflects its cosmetics-primary identity rather than its skincare-extension identity. The borderline classifications in pre-reg §3.4 were prudent.

**LLM panel temporal validity.** The six-slot reference panel reflects model versions in market as of mid-2026. Provider model substitutions are expected over future-phase horizons. v0.20's Type 2 cases should be interpreted as the pattern observed against the locked panel at acquisition time, not as a permanent property of the substrate. Re-acquisition with a future panel would test the temporal stability of the dissociation signatures.

**Cultural channel definition is substrate-specific.** v0.20's  $R_{cult}$  battery (popular, celebrity, viral / cult-favorite) is calibrated to skincare's consumer discovery surface. Other substrates may require different cultural-channel frame wording — fragrance's "perfumistas-native" lexicon (per v0.18); audiophile's "audiophile-community" lexicon (per v0.19); skincare's "celebrity-influencer" lexicon (per v0.20). A general specification of the cultural channel that abstracts above substrate-specific lexicons is a Protocol v1.6 candidate.

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## 5. DEVIATIONS

The DEVIATIONS protocol carries forward from v0.16 through v0.19. One entry is open ex-ante; no additional entries were opened during acquisition.

**Entry 0 — Pre-acquisition COI screen** (2026-05-21, ex-ante). Per the AIAS™ Declarations standard. Screen subject: Samsung Electronics America (author’s primary employer) potential exposure to v0.20 registry. Screen result: none of the 24 brands in the locked registry is affiliated with Samsung. Samsung’s historical beauty exposure was via Cheil Industries (spun off pre-2026 in corporate restructuring); no current overlap with the v0.20 registry. The author’s role at Samsung carries no commercial, strategic, or signaling interest in any v0.20 brand outcome. Disposition: no deviation from pre-registration warranted. Lock state: v0.20-prereg-r1 unchanged.

No deviations were opened during Phase A or Phase B acquisition.

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## 6. Conclusion

v0.20 produces four pre-registered hypothesis verdicts on the AIAS™ Presence Measurement Protocol’s first prospective phase under v1.5: H\_Type2\_emergence PARTIAL, H\_Regime4\_skincare FALSIFIED, H\_Dissociation\_substrate\_generalization GENERALIZED, and H\_IdentityLoad\_moderator (4-leg joint) NARROWED.

**For the substantive construct.** H\_Type2\_emergence’s PARTIAL verdict populates the v1.5 dissociation framework’s third quadrant for the first time. Glossier, Rhode, and Augustinus Bader are the empirical anchors. The cultural-channel Recall pathway operates as a distinct mechanism from canonical-channel Recall — the brands that an LLM lists when asked “what skincare brands are popular right now” are not the brands an LLM lists when asked “what skincare brands do dermatologists recommend,” and the brand-by-brand asymmetry between those frames is the v1.5 two-channel decomposition’s empirical signal.

**For the canonical methodology.** H\_Dissociation\_substrate\_generalization’s GENERALIZED verdict extends the multi-component construct’s empirical anchor base to four substrate families with multi-cell distribution in v0.20. For AIAS™ 1.0’s headline cross-substrate generalization claim, the construct is now defensibly anchored. H\_Regime4’s FALSIFIED verdict supplies the first prospective evidence of v1.5 C2’s behavior — the rule operates as the methodology paper specified: variance-rich cells clear, saturated cells reject, bi-

modal cells reject. The rule's prospective utility is established.

**For the moderator hypothesis.** The NARROWED joint verdict on H\_IdentityLoad\_moderator coexists with a substantive channel-asymmetry signal in v0.20's raw data that exactly tracks the IL gradient (Cell C R\_cat-dominant, Cell B R\_cult-dominant). The tension is itself informative: the joint matrix's NARROWED routing is structurally driven by Regime 4's C2 failure shapes, not by the moderator's substantive non-operation. Future v1.5 refinements may benefit from a moderator pathway evaluable independently of Regime 4's conditions.

**For v0.21 and beyond.** v0.20's verdicts point at three productive directions. The Type 2 PARTIAL verdict invites substrates with even higher Cell B IL saturation that might produce EMERGED-level Type 2 counts. The Cell C saturation finding invites finer-grained Cell C stratification — separating icon-tier from mainstream-tier mass-prestige — to surface within-cell variance for Phase D  $\rho$  analysis. The IL moderator joint-verdict / channel-asymmetry tension invites a Protocol v1.6 candidate: a direct moderator-evaluation pathway independent of Regime 4. Each direction is pre-registerable; each leaves v0.20's results unchanged.

The pre-registration discipline applied across the v0.20 lifecycle — locking the v1.5 thresholds prospectively for the first time; documenting Cell C cascade-saturation as the substantive realization of a pre-reg §8 prediction; carrying the four-hypothesis verdict matrix forward unaltered through acquisition — is the protocol-level commitment that distinguishes the AIAS™ program from looser AI-visibility tooling. The verdicts are what the data produces. The construct, the moderator, and the substrate together carry more structure than any single phase can resolve.

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## Declarations

**Declaration of interest.** The author is employed by Samsung Electronics America (Director, Corporate Brand Creative and Governance). The AIAS™ Presence Measurement Protocol and the work reported here are the author's independent academic research, conducted outside the scope of employment, in the author's role as faculty at the School of Visual Arts MPS Branding Program and founder of Third System™. Samsung had no role in the design, execution, analysis, or interpretation of this work. None of the 24 brands in the v0.20 registry is affiliated with Samsung Electronics America (pre-acquisition COI screen documented as DEVIATIONS Entry 0).

**Funder.** Self-funded.

**Ethics.** Not applicable; no human subjects. The research uses publicly accessible LLM APIs queried with non-personal, category-anchored prompts.

**Data and code availability.** All pre-registration artifacts, acquisition data, scoring code, and verdict outputs are deposited under Open Science Framework project ec6wh at [osf.io/ec6wh/v20/](https://osf.io/ec6wh/v20/). The pre-registration is locked at commit 4e5ab60 on git tag v0.20-prereg-r1, branch v0.20-skincare-1l-gradient. The acquisition is locked at git tag v0.20-acquisition-locked. All four hypothesis verdict matrices are specified ex-ante in `prereg/v0_20_registry.json`.

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